Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code S - 0001 - WLA		
Kurskod / Course code DT171G	Provkod/ Test code T101	Tentamensdatum / Examination date 2024-08-23
Kursnamn / Course name Datateknik GR (B), Inbäddade system för Internet of Things		
Provnamn / Test name Skriftlig salstentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	4		16			
2	X	1,5		17			
3	X	3,5		18			
4	X	2		19			
5	~~X~~			20			
6	X	4,5		21			
7	X	10		22			
8				23			
9				24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		25,5	Betyg / Grade D	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted 11	**Inlämnad tentamen / Submitted exam** 0	**Leg kontroll / Control identification** ✓	**Sign. tentamensvakt / Sign. invigilator**

2707183

Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-08-21 kl. 12:44:23

Mittuniversitetet
MID SWEDEN UNIVERSITY

Anonymitetskod/ Anonymus code: S-0001-WLA

Sidnr / Page no: 1

Lärarens anteckning / Teachers note:

Uppgift 1

Given:

$R_1 = 350\,\Omega$

$R_2 = 200\,\Omega$

$I = 40\,mA = 0{,}04\,A$

R1 and R2 are in parallel which means

$$\frac{1}{R_{(total)}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{(total)}} = \frac{1}{350} + \frac{1}{200}$$

$$\frac{1}{R} = \frac{4}{1700} + \frac{7}{1400}$$

$$\frac{1}{R} = \frac{11}{1400}$$

$$11R = 1400$$

$$R = \frac{1400}{11}$$

$$R = 127{,}27\,\Omega$$

Using Ohm's Law:

$U = I\,R$

$U = 0{,}04 \cdot 127.27\,\Omega$

$U = \boxed{5{,}09\,V}$

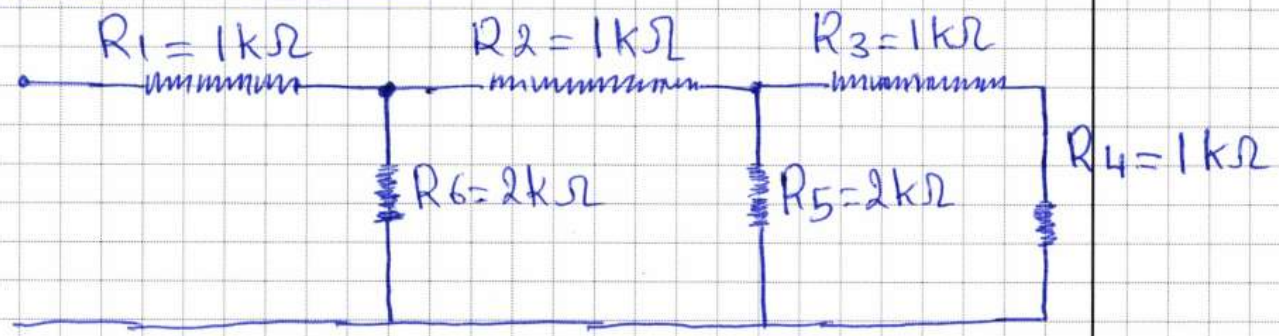
4

Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

Uppgif (2)

Initial circuit:

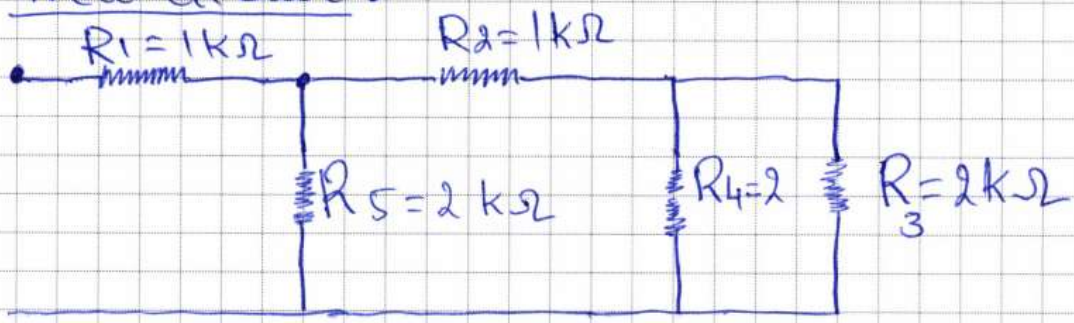


R_3 and R_4 and in series

$R_{(total)} = R_3 + R_4$

$= 1 + 1 = \boxed{2\,k\Omega}$

New circuit:



R3 and R4 are in parallel

$R_{(total)} = \frac{1}{R_3} + \frac{1}{R_4}$

$R_{(total)} = \frac{1}{2} + \frac{1}{2}$

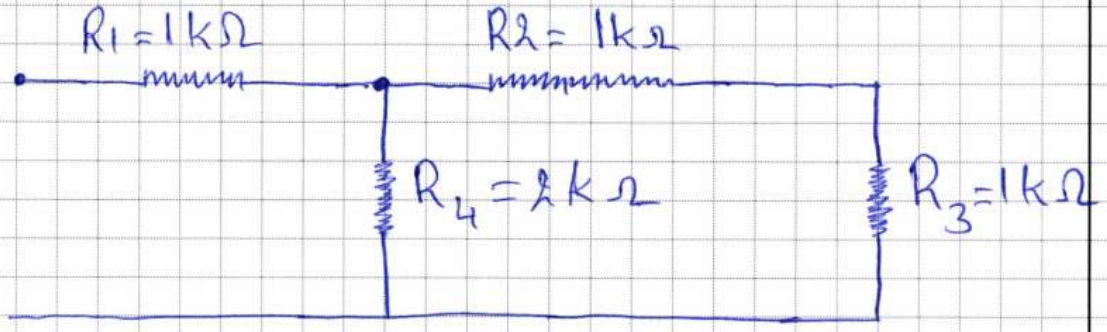
$R_{(total)} = \frac{2}{2}$

$R_{total} = \boxed{1k\Omega}$

Lärarens anteckning / Teachers note:

Uppgift 2 (Fortsättning 1)

New circuit:



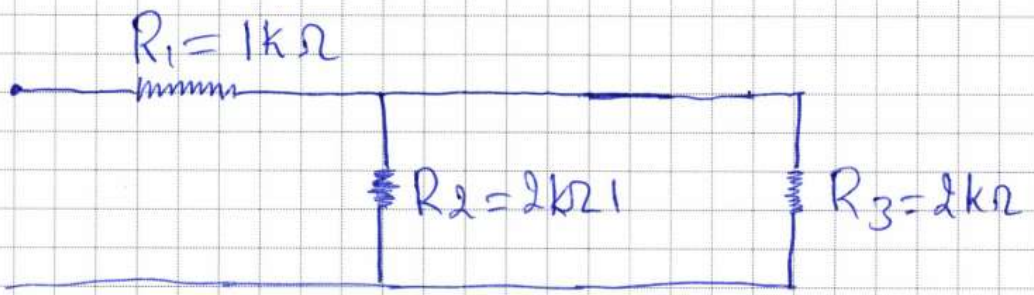
R_2 and R_3 are in series

$R_{(total)} = R_2 + R_3$

$R_{(total)} = 1 + 1$

$= \boxed{2\ k\Omega}$

New circuit



R_2 and R_3 are in parallel

$R_{(total)} = \frac{1}{R_2} + \frac{1}{R_3}$

$= \frac{1}{2} + \frac{1}{2}$

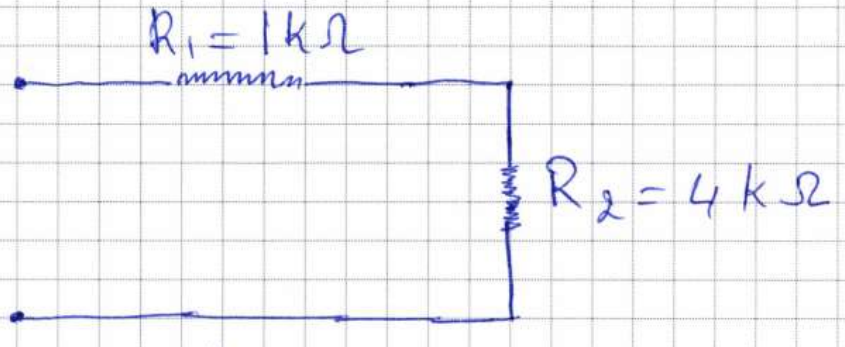
$= \frac{2}{2}$

$= \boxed{2\ k\Omega}$

Lärarens anteckning / Teachers note:

Uppgift 2 (fortsättning 2)

New circuit:



R_1 and R_2 are in series

$R_{(total)} = R_1 + R_2$

$= 1 + 5$

$= \boxed{6k\Omega}$

Svar:

$R_{(ab)} = \boxed{6k\Omega}$ ✓

1,5

Mittuniversitetet
MID SWEDEN UNIVERSITY

Anonymitetskod/ Anonymus code: S-0001-WLA

Sidnr / Page no: 5

Lärarens anteckning / Teachers note:

Uppgift (3)

- a sensor is device that captures and transforms physical real world phenomenas into an electrical signal that can be used for control and analysis.

Exemple: temperature sensor

a temperatur sensor transforms temperature into an analog singnal that can be used for different purposes.

- An actuator Transforms the input it recieves into mechanical motion that affects the pysical world.

Example: Servomotor

servomotors transforms electrical signal into mechanical motion causing movement in a robotic arm.

Skriv ej på baksidan! Do not write on the other side!

Mittuniversitetet
MID SWEDEN UNIVERSITY

Anonymitetskod/ Anonymus code: S-0001-WLA

Sidnr / Page no: 6

Lärarens anteckning / Teachers note:

Uppgift (4)

Problem 1: Signal Noise:

analog sensors can pick up noise leading to inaccurate readings

problem 2: Signal Resolution
the microcontroller's ADC (Analog to digital converter) resolution may be low leading to the sensor not picking up details.

Signal levels!

2

Mittuniversitetet
MID SWEDEN UNIVERSITY

Anonymitetskod/ Anonymus code: S-0001-WLA

Sidnr / Page no: 7

Lärarens anteckning / Teachers note:

Uppgift (6)

UART (Universall asynchrounous Reciever / transmitter)
UART is a serial asynchrounous communication protokoll used for short distance communication between embedded systems. it uses two pins one for sending data (TX) and one for recieving data (RX).
UART operates on a master slave configuration. the master (sender) send data to the slave (reciever). Start and stop bits frame each bit providing synchronization.

UART is used in senarios like connecting a microcontroller to a sensor for example as in GPS modules.

4,5

Lärarens anteckning / Teachers note:

Uppgif (7) (components and connection)

Components:

1 Arduino Uno

1 sensor TMP 36

3 Led lights

3 resistors

Connection:

Step (1) the sensor:

the sensor has three legs: Vout, Vs; GND

connect Vout to pin A0 and then connect Vs to 5V and lastly connect GND to ground

Step (2) the led lights:

Connect the anode (long leg) of the blue led light to pin 2 and then connect the cathode (short leg) to one leg of a 220 ohm resistor and lastly connect the other leg of the resistor to ground.

do the same thing with the green and red led lights: anode to pin 3 for the green led and cathode to one side of a resistor that goes to ground.

Lastly pin 4 for the red Led and same procedure.

Lärarens anteckning / Teachers note:

Uppgift (7) (Programkod)

```
const int sensorPin = A0;
const int blueLed = 2;
const int greenLed = 3;
const int redLed = 4;
void setup(){
    pinMode(2, OUTPUT);
    pinMode(3, OUTPUT);
    pinMode(4, OUTPUT);
    Serial.begin(9600);
}

float readTemperature(intPin){
    int reading = analogRead(pin);
    float voltage = reading * (5.0 / 1024.0);
    float temperature = (voltage - 0,5) * 100;
    return temperature;
}
```

Lärarens anteckning / Teachers note:

Uppgift 7 (programkod fortsättning)

```
void updateLed(float temperature){
    if(temperature < -22){
        digitalWrite(2, HIGH);
        digitalWrite(3, LOW);
        digitalWrite(4, LOW);
    }
    else if(temperature > -18){
        digitalWrite(2, LOW);
        digitalWrite(3, LOW);
        digitalWrite(4; HIGH);
    }
    else{
        digitalWrite(2, LOW);
        digitalWrite(3; HIGH);
        digitalWrite(4; LOW);
}
```

Flytta!

Mittuniversitetet
MID SWEDEN UNIVERSITY

Anonymitetskod/ Anonymus code: S - 0 0 0 1 - W L A

Sidnr / Page no: 11

Lärarens anteckning / Teachers note:

Uppgift 7 (programkod fortsättning 2)

```
void loop() {
    float temperature = readTemperature(A0);
    Serial.println(temperature);
    UpdateLed(temperature);

}
```

10

Skriv ej på baksidan! Do not write on the other side!